

CO2 Emissions & Income Groups Predictors

Source disclosure

Reference sources for the techniques used:

- Visualization: ggplot2 reference for `geom_jitter`, https://ggplot2.tidyverse.org/reference/geom_jitter.html
- Cleaning: R for Data Science, data tidying chapter, <https://r4ds.hadley.nz/data-tidy.html>
- Other: R Markdown Cookbook, tables with `kable`, <https://bookdown.org/yihui/rmarkdown-cookbook/kable.html>

Libraries

```
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.3

Import the data files

Each file is in wide format with one row per country and one column per year.

```
sanitation <- read_csv("../data/predictors/at_least_basic_sanitation_overall_access_percent.csv")
fertility <- read_csv("../data/predictors/children_per_woman_total_fertility.csv")
co2 <- read_csv("../data/predictors/co2_pcap_cons.csv")
happiness <- read_csv("../data/predictors/hapiscore_whr.csv")
income <- read_csv("../data/predictors/ilevels4_wb.csv", col_types = cols(.default = "c"))
poverty <- read_csv("../data/predictors/number_of_people_in_poverty.csv")
```

Dropping Year Columns

The year columns from 2023 onwards are dropped first. Each dataset is then changed to long format so that all of them share the same geo, name and year keys, which avoids duplicated year column names when joining.

```
drop_and_pivot <- function(df, value_name) {
  df |>
    select(-any_of(as.character(2023:2100))) |>
    pivot_longer(
      cols = -c(geo, name),
      names_to = "year",
      values_to = value_name
    ) |>
    mutate(year = as.integer(year))
}

sanitation_long <- drop_and_pivot(sanitation, "sanitation_access_pct")
fertility_long <- drop_and_pivot(fertility, "children_per_woman")
co2_long <- drop_and_pivot(co2, "co2_per_capita")
happiness_long <- drop_and_pivot(happiness, "happiness_score")
income_long <- drop_and_pivot(income, "income_level")
poverty_long <- drop_and_pivot(poverty, "people_in_poverty")
```

Join into a single dataset

```
combined <- list(
  sanitation_long, fertility_long, co2_long,
  happiness_long, income_long, poverty_long
) |>
  reduce(full_join, by = c("geo", "name", "year")) |>
  arrange(geo, year)

glimpse(combined)
```

Rows: 43,485

Columns: 9

```
$ geo      <chr> "afg", "afg", "afg", "afg", "afg", "afg", "afg", ~
$ name     <chr> "Afghanistan", "Afghanistan", "Afghanistan", "Af~
$ year     <int> 1800, 1801, 1802, 1803, 1804, 1805, 1806, 1807, ~
```

```
$ sanitation_access_pct <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ children_per_woman    <dbl> 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, ~
$ co2_per_capita        <dbl> 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, ~
$ happiness_score       <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ income_level          <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ people_in_poverty     <dbl> 3082964, 3082964, 3082964, 3082964, 3082964, 308~
```

```
range(combined$year)
```

```
[1] 1800 2022
```

```
n_distinct(combined$geo)
```

```
[1] 195
```

```
head(combined)
```

```
# A tibble: 6 x 9
  geo   name      year sanitation_access_pct children_per_woman co2_per_capita
  <chr> <chr>    <int>          <dbl>          <dbl>          <dbl>
1 afg  Afghanist~ 1800             NA             7             0.001
2 afg  Afghanist~ 1801             NA             7             0.001
3 afg  Afghanist~ 1802             NA             7             0.001
4 afg  Afghanist~ 1803             NA             7             0.001
5 afg  Afghanist~ 1804             NA             7             0.001
6 afg  Afghanist~ 1805             NA             7             0.001
# i 3 more variables: happiness_score <dbl>, income_level <chr>,
#   people_in_poverty <dbl>
```

Ian Peralta: CO2 Emissions and Income Groups

Population, the response, is not yet in `combined`, so it is imported here.

```
population      <- read_csv("../data/response/pop.csv")
population_long <- drop_and_pivot(population, "population")
```

CO2 Emissions per Capita: Consumption-Based

co2_per_capita is quantitative and continuous, measured in tonnes of CO2 per person per year. Values run from near zero to several tens of tonnes.

```
co2_long |>
  summarise(
    n          = n(),
    n_missing  = sum(is.na(co2_per_capita)),
    pct_missing = round(100 * mean(is.na(co2_per_capita)), 1),
    first_year = min(year[!is.na(co2_per_capita)]),
    last_year  = max(year[!is.na(co2_per_capita)])
  ) |>
  knitr::kable()
```

n	n_missing	pct_missing	first_year	last_year
42816	1805	4.2	1800	2022

About 4% of records are missing. Most of this is structural, since many countries are only reported from later years.

```
co2_long |>
  filter(co2_per_capita < 0) |>
  arrange(co2_per_capita) |>
  knitr::kable(digits = 3)
```

geo	name	year	co2_per_capita
pan	Panama	2012	-0.579
pan	Panama	2005	-0.511
pan	Panama	2013	-0.325
pan	Panama	2003	-0.107
pan	Panama	2004	-0.085
pan	Panama	2006	-0.079
pan	Panama	2011	-0.004

Seven values from Panama are slightly negative. Consumption-based accounting subtracts exported emissions, so a trade hub can go net negative. These are kept as a real feature of the method, but they matter for modeling.

```
co2_latest <- co2_long |>
  filter(!is.na(co2_per_capita), year == max(year[!is.na(co2_per_capita)]))

co2_long |>
  filter(!is.na(co2_per_capita)) |>
  summarise(mean = mean(co2_per_capita), median = median(co2_per_capita),
            sd = sd(co2_per_capita), min = min(co2_per_capita),
            max = max(co2_per_capita)) |>
  knitr::kable(digits = 2, caption = "CO2 per capita, all years")
```

Table 1: CO2 per capita, all years

mean	median	sd	min	max
1.65	0.12	3.98	-0.58	62.15

```
co2_latest |>
  summarise(year = unique(year), n_countries = n(), mean = mean(co2_per_capita),
            median = median(co2_per_capita), sd = sd(co2_per_capita),
            min = min(co2_per_capita), max = max(co2_per_capita)) |>
  knitr::kable(digits = 2, caption = "CO2 per capita, 2022")
```

Table 2: CO2 per capita, 2022

year	n_countries	mean	median	sd	min	max
2022	192	4.71	2.78	5.33	0.03	27.26

```
co2_long |>
  filter(!is.na(co2_per_capita)) |>
  group_by(year) |>
  summarise(mean_co2 = mean(co2_per_capita)) |>
  ggplot(aes(year, mean_co2)) +
  geom_line(color = "steelblue", linewidth = 0.8) +
  labs(x = "Year", y = "Mean CO2 per capita, tonnes") +
  theme_minimal()
```

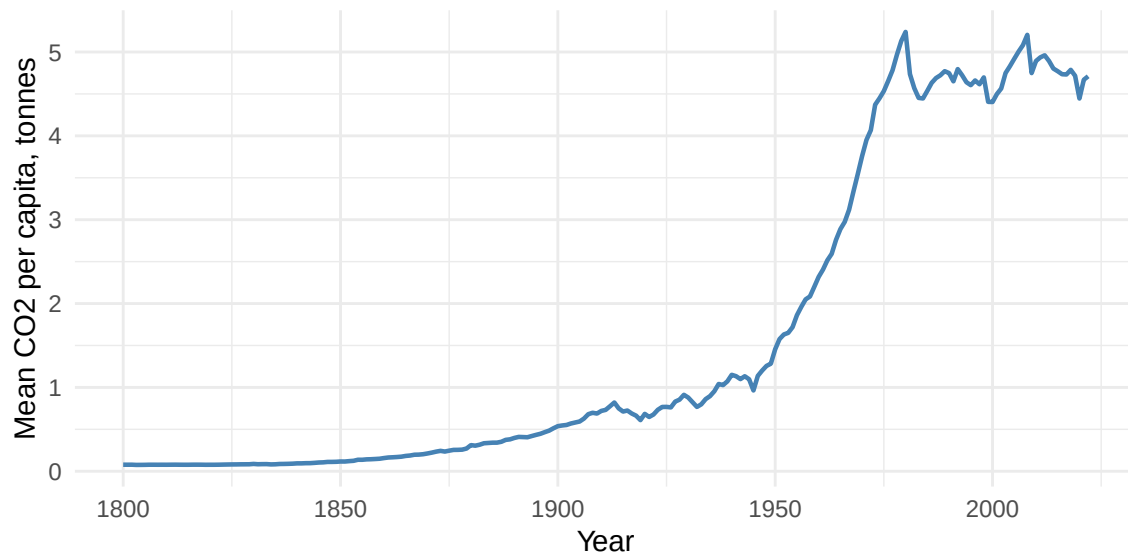


Figure 1: Mean CO2 emissions per capita by year

```
ggplot(co2_latest, aes(co2_per_capita)) +
  geom_histogram(bins = 25, fill = "blue", color = "white") +
  labs(x = "CO2 per capita, tonnes", y = "Number of countries") +
  theme_minimal()
```

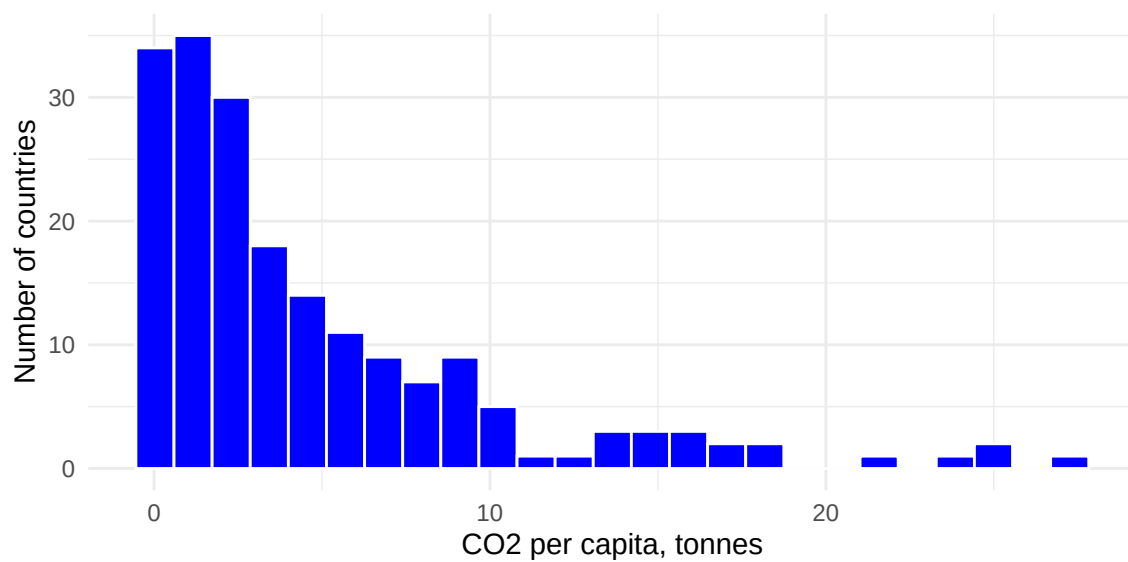


Figure 2: Distribution of CO2 emissions per capita, 2022

```
bind_rows(
  co2_latest |> slice_max(co2_per_capita, n = 5) |> mutate(group = "Highest"),
  co2_latest |> slice_min(co2_per_capita, n = 5) |> mutate(group = "Lowest")
) |>
select(group, name, co2_per_capita) |>
knitr::kable(digits = 2, caption = "Highest and lowest CO2 per capita, 2022")
```

Table 3: Highest and lowest CO2 per capita, 2022

group	name	co2_per_capita
Highest	Qatar	27.26
Highest	Singapore	25.43
Highest	Brunei	24.48
Highest	Kuwait	24.01
Highest	UAE	21.12
Lowest	Congo, Dem. Rep.	0.03
Lowest	Burundi	0.05
Lowest	Somalia	0.05
Lowest	Central African Republic	0.07
Lowest	Chad	0.07

The 2022 distribution is right-skewed, with most countries below 5 tonnes. The highest are Qatar, Singapore, Brunei, Kuwait and the UAE, small wealthy economies tied to oil, gas or trade. The lowest are the DR Congo, Burundi, Somalia, the Central African Republic and Chad. Average emissions climbed through industrialization and have leveled off since about 1980.

```
co2_pop_2022 <- co2_latest |>
  inner_join(population_long |> filter(year == 2022),
    by = c("geo", "name", "year"))

cor(co2_pop_2022$co2_per_capita, log10(co2_pop_2022$population))
```

```
[1] -0.0615751
```

```
ggplot(co2_pop_2022, aes(population, co2_per_capita)) +
  geom_point(alpha = 0.6, color = "blue") +
  scale_x_log10(labels = scales::comma) +
  labs(x = "Population, log scale", y = "CO2 per capita, tonnes") +
  theme_minimal()
```

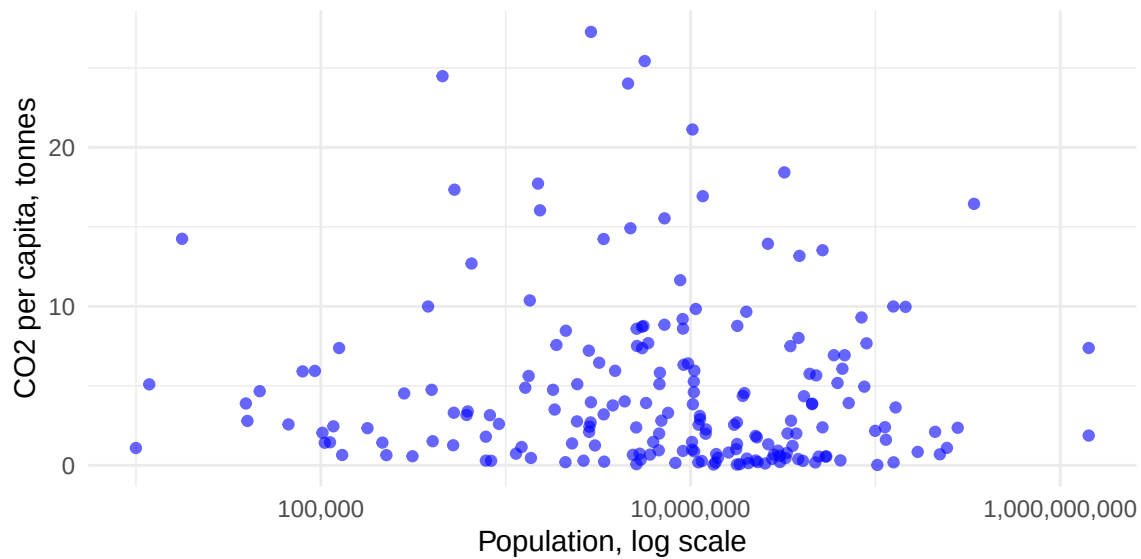


Figure 3: CO2 emissions per capita against population, 2022

There is almost no relationship between population and per-capita emissions, with $r = -0.06$ against log population. This is expected, since the measure is already per person and depends more on income and energy use than country size.

4 Income Groups: World Bank

`income_level` is categorical and ordinal with four levels: Low, Lower middle, Upper middle and High income, assigned by the World Bank from GNI per capita.

```
income_long <- income_long |>
  mutate(income_level = factor(
    income_level,
    levels = c("Low income", "Lower middle income",
               "Upper middle income", "High income"),
    ordered = TRUE
  ))

income_long |>
  summarise(
    n = n(),
    n_missing = sum(is.na(income_level)),
    pct_missing = round(100 * mean(is.na(income_level)), 1),
    first_year = min(year[!is.na(income_level)]),
```

```

    last_year = max(year[!is.na(income_level)])
  ) |>
  knitr::kable()

```

n	n_missing	pct_missing	first_year	last_year
6984	0	0	1987	2022

There is no missing data from 1987 to 2022, the years the World Bank classifies. Earlier years will appear missing once this is joined with longer variables, which is structural.

```

income_latest <- income_long |>
  filter(!is.na(income_level), year == max(year[!is.na(income_level)]))

income_latest |>
  count(income_level) |>
  mutate(pct = round(100 * n / sum(n), 1)) |>
  knitr::kable(caption = "World Bank income group, 2022")

```

Table 4: World Bank income group, 2022

income_level	n	pct
Low income	26	13.4
Lower middle income	55	28.4
Upper middle income	53	27.3
High income	60	30.9

```

ggplot(income_latest, aes(income_level)) +
  geom_bar(fill = "orange") +
  scale_x_discrete(labels = scales::label_wrap(12)) +
  labs(x = "Income group", y = "Number of countries") +
  theme_minimal()

```

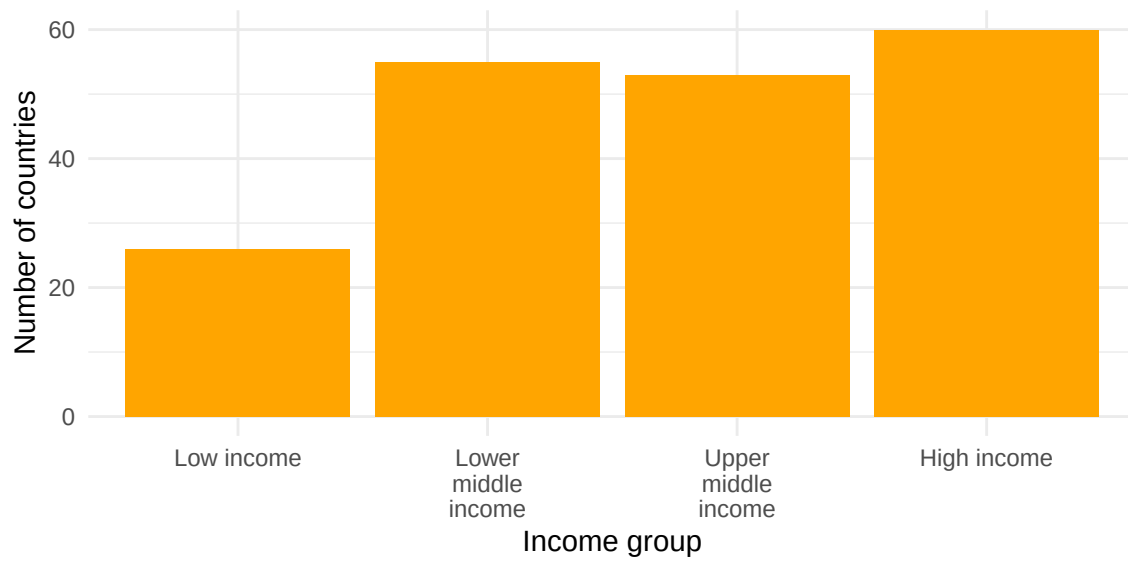


Figure 4: Number of countries by income group, 2022

```
income_long |>
  filter(!is.na(income_level)) |>
  count(year, income_level) |>
  group_by(year) |>
  mutate(pct = n / sum(n)) |>
  ggplot(aes(year, pct, fill = income_level)) +
  geom_area() +
  labs(x = "Year", y = "Share of countries", fill = "Income group") +
  theme_minimal()
```

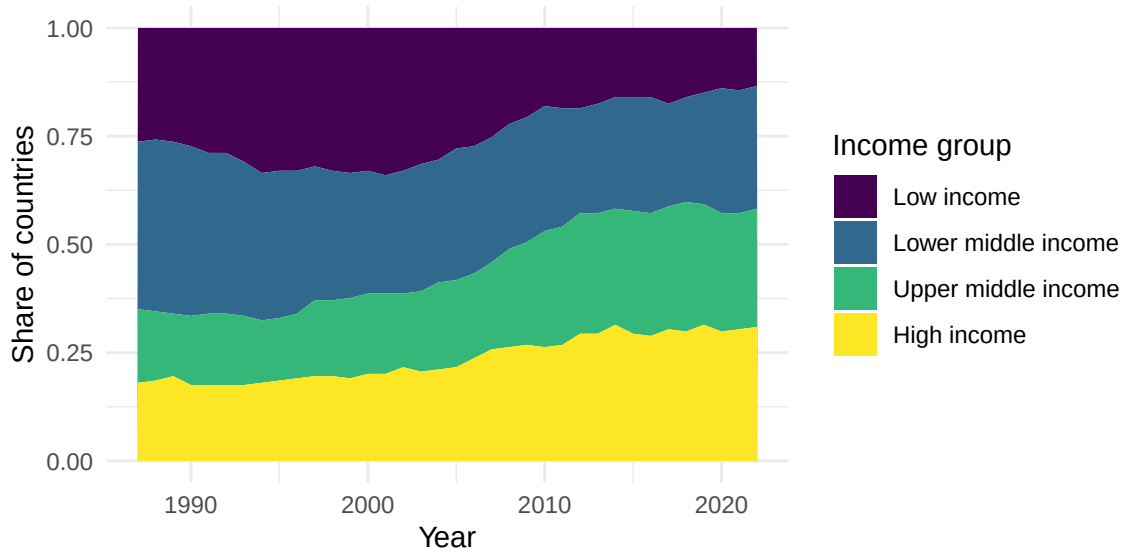


Figure 5: Share of countries in each income group, 1987 to 2022

In 2022 High income is the largest group with 60 countries, or 30.9%, followed by Lower middle at 55, Upper middle at 53 and Low income at 26. Since 1987 the High income share rose from 18% to almost 31%, while Low income fell from 26% to 13%.

```
income_pop_2022 <- income_latest |>
  inner_join(population_long |> filter(year == 2022),
    by = c("geo", "name", "year"))

income_pop_2022 |>
  group_by(income_level) |>
  summarise(n = n(), median_pop = median(population), mean_pop = mean(population)) |>
  knitr::kable(digits = 0, caption = "Population by income group, 2022")
```

Table 5: Population by income group, 2022

income_level	n	median_pop	mean_pop
Low income	26	21515450	27571761
Lower middle income	55	13759501	58989892
Upper middle income	53	5305270	52887214
High income	60	5294268	20262448

Population by income group, 2022

```
ggplot(income_pop_2022, aes(income_level, population)) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.6, color = "orange", size = 2) +
  stat_summary(fun = median, geom = "crossbar", width = 0.5,
              color = "black", linewidth = 0.4) +
  scale_x_discrete(labels = scales::label_wrap(12)) +
  scale_y_log10(labels = scales::comma) +
  labs(x = "Income group", y = "Population, log scale") +
  theme_minimal()
```

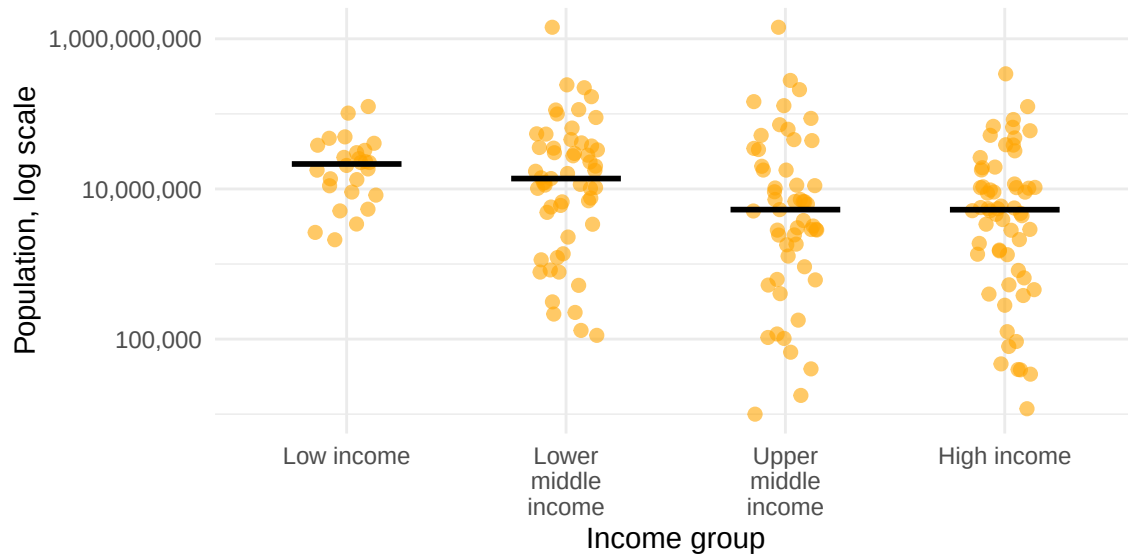


Figure 6: Population by income group, 2022

Median population is highest in Low income at 21.5 million and lowest in High income at 5.3 million, and every group mean exceeds its median. In Lower middle income the mean of 59 million is over four times the median of 13.8 million, driven by India and Pakistan. China and Indonesia do the same in Upper middle income. Population size and income group look largely unrelated.